분산시스템 과제 #10

박종현, 2025-06-07
공과대학 컴퓨터정보통신공학과

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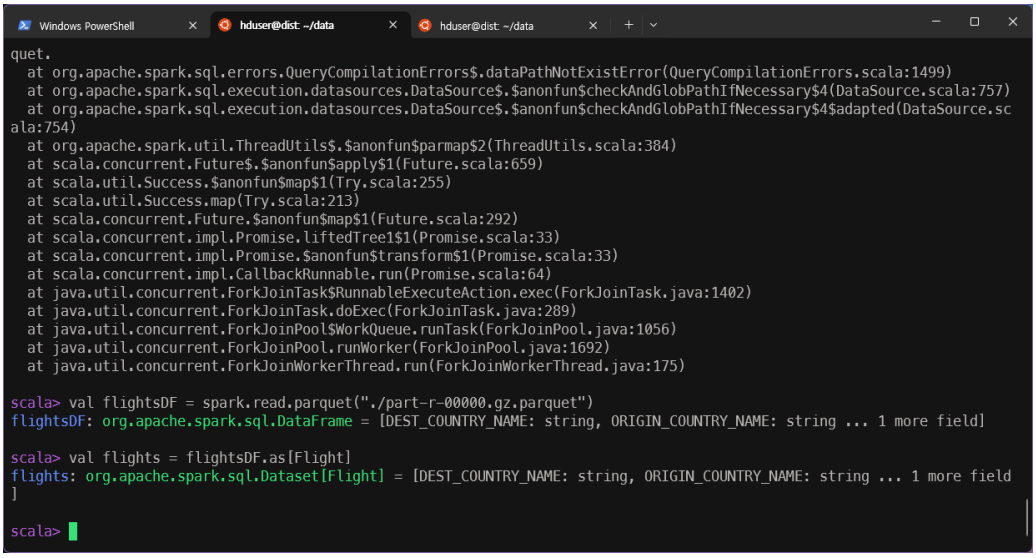
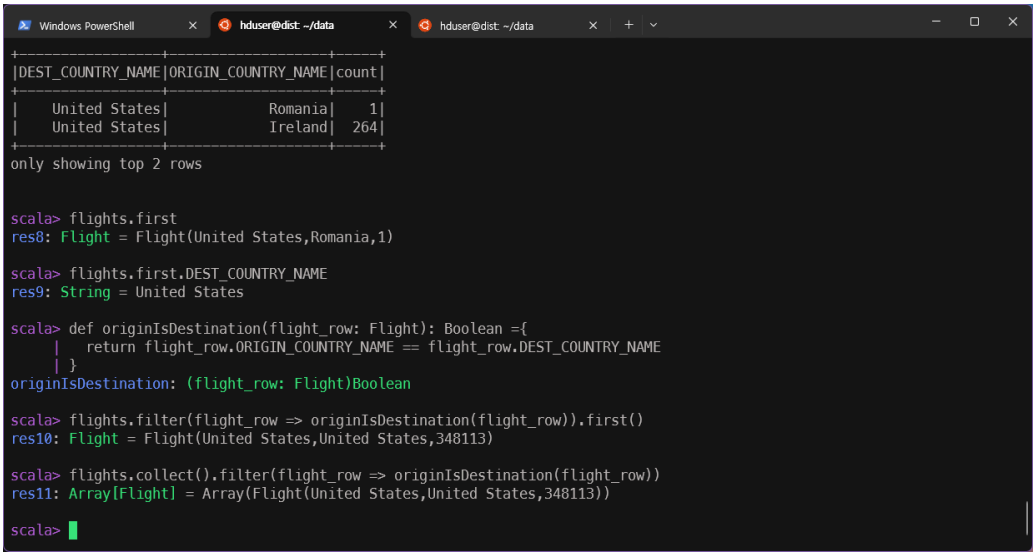


```
quet.
  at org.apache.spark.sql.errors.QueryCompilationErrors$.dataPathNotExistError(QueryCompilationErrors.scala:1499)
  at org.apache.spark.sql.execution.datasources.DataSource$.$anonfun$checkAndGlobPathIfNecessary$4(DataSource.scala:757)
  at org.apache.spark.sql.execution.datasources.DataSource$.$anonfun$checkAndGlobPathIfNecessary$4$adapted(DataSource.sc
ala:754)
  at org.apache.spark.util.ThreadUtils$.$anonfun$parmap$2(ThreadUtils.scala:384)
  at scala.concurrent.Future$.$anonfun$apply$1(Future.scala:659)
  at scala.util.Success.$anonfun$map$1(Try.scala:255)
  at scala.util.Success.map(Try.scala:213)
  at scala.concurrent.Future.$anonfun$map$1(Future.scala:292)
  at scala.concurrent.impl.Promise.liftedTree1$1(Promise.scala:33)
  at scala.concurrent.impl.Promise.$anonfun$transform$1(Promise.scala:33)
  at scala.concurrent.impl.CallbackRunnable.run(Promise.scala:64)
  at java.util.concurrent.ForkJoinTask$RunnableExecuteAction.exec(ForkJoinTask.java:1402)
  at java.util.concurrent.ForkJoinTask.doExec(ForkJoinTask.java:289)
  at java.util.concurrent.ForkJoinPool$WorkQueue.runTask(ForkJoinPool.java:1056)
  at java.util.concurrent.ForkJoinPool.runWorker(ForkJoinPool.java:1692)
  at java.util.concurrent.ForkJoinWorkerThread.run(ForkJoinWorkerThread.java:175)

scala> val flightsDF = spark.read.parquet("./part-r-00000.gz.parquet")
flightsDF: org.apache.spark.sql.DataFrame = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ... 1 more field]

scala> val flights = flightsDF.as[Flight]
flights: org.apache.spark.sql.Dataset[Flight] = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ... 1 more field
]

scala>
```



```
+-----------------+-------------------+-----+
|DEST_COUNTRY_NAME|ORIGIN_COUNTRY_NAME|count|
+-----------------+-------------------+-----+
|    United States|            Romania|    1|
|    United States|            Ireland|  264|
+-----------------+-------------------+-----+
only showing top 2 rows


scala> flights.first
res8: Flight = Flight(United States,Romania,1)

scala> flights.first.DEST_COUNTRY_NAME
res9: String = United States

scala> def originIsDestination(flight_row: Flight): Boolean ={
     |   return flight_row.ORIGIN_COUNTRY_NAME == flight_row.DEST_COUNTRY_NAME
     | }
originIsDestination: (flight_row: Flight)Boolean

scala> flights.filter(flight_row => originIsDestination(flight_row)).first()
res10: Flight = Flight(United States,United States,348113)

scala> flights.collect().filter(flight_row => originIsDestination(flight_row))
res11: Array[Flight] = Array(Flight(United States,United States,348113))

scala>
```

```
Windows PowerShell   hduser@dist: ~/data   hduser@dist: ~/data
  at java.util.concurrent.ForkJoinPool.runWorker(ForkJoinPool.java:1692)
  at java.util.concurrent.ForkJoinWorkerThread.run(ForkJoinWorkerThread.java:175)

scala> val flightsDF = spark.read.parquet("./part-r-00000.gz.parquet")
flightsDF: org.apache.spark.sql.DataFrame = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ... 1 more field]

scala> val flights = flightsDF.as[Flight]
flights: org.apache.spark.sql.Dataset[Flight] = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ... 1 more field
]

scala> flights.show(2)
+-----------------+-------------------+-----+
|DEST_COUNTRY_NAME|ORIGIN_COUNTRY_NAME|count|
+-----------------+-------------------+-----+
|    United States|            Romania|    1|
|    United States|            Ireland|  264|
+-----------------+-------------------+-----+
only showing top 2 rows


scala> flights.first
res8: Flight = Flight(United States,Romania,1)

scala> flights.first.DEST_COUNTRY_NAME
res9: String = United States

scala>
```

```
Windows PowerShell   hduser@dist: ~/data   hduser@dist: ~/data

scala> flights.first
res8: Flight = Flight(United States,Romania,1)

scala> flights.first.DEST_COUNTRY_NAME
res9: String = United States

scala> def originIsDestination(flight_row: Flight): Boolean ={
     |   return flight_row.ORIGIN_COUNTRY_NAME == flight_row.DEST_COUNTRY_NAME
     | }
originIsDestination: (flight_row: Flight)Boolean

scala> flights.filter(flight_row => originIsDestination(flight_row)).first()
res10: Flight = Flight(United States,United States,348113)

scala> flights.collect().filter(flight_row => originIsDestination(flight_row))
res11: Array[Flight] = Array(Flight(United States,United States,348113))

scala> val destinations = flights.map(f => f.DEST_COUNTRY_NAME)
destinations: org.apache.spark.sql.Dataset[String] = [value: string]

scala> val localDestinations = destinations.take(5)
localDestinations: Array[String] = Array(United States, United States, United States, Egypt, Equatorial Guinea)

scala> case class FlightMetadata(count: BigInt, randomData: BigInt)
defined class FlightMetadata
```

```
Windows PowerShell   hduser@dist: ~/data   hduser@dist: ~/data
scala> val flightsMeta = spark.range(500).map(x => (x, scala.utilRandom.nextLong)).withColumnRenamed("_1","count").withC
olumnRenamed("_2","randomData").as[FlightMetadata]
<console>:24: error: object utilRandom is not a member of package scala
       val flightsMeta = spark.range(500).map(x => (x, scala.utilRandom.nextLong)).withColumnRenamed("_1","count").withC
olumnRenamed("_2","randomData").as[FlightMetadata]
                                                  ^

scala> val flightsMeta = spark.range(500).map(x => (x, scala.util.Random.nextLong)).withColumnRenamed("_1","count").with
ColumnRenamed("_2","randomData").as[FlightMetadata]
flightsMeta: org.apache.spark.sql.Dataset[FlightMetadata] = [count: bigint, randomData: bigint]

scala> val flights2 = flights.joinWith(flightsMeta, flights.col("count") === flightsMeta.col("count"))
flights2: org.apache.spark.sql.Dataset[(Flight, FlightMetadata)] = [_1: struct<DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY
_NAME: string ... 1 more field>, _2: struct<count: bigint, randomData: bigint>]

scala> flights2.take(2)
res12: Array[(Flight, FlightMetadata)] = Array((Flight(United States,Uganda,1),FlightMetadata(1,2764182393313317269)), (
Flight(United States,French Guiana,1),FlightMetadata(1,2764182393313317269)))

scala> val flights2 = flights.join(flightsMeta, Seq("count"))
flights2: org.apache.spark.sql.DataFrame = [count: bigint, DEST_COUNTRY_NAME: string ... 2 more fields]

scala> flights2.take(2)
res13: Array[org.apache.spark.sql.Row] = Array([1,United States,Uganda,5767147658854483195], [1,United States,French Gui
ana,5767147658854483195])

scala>
```

```
Windows PowerShell   hduser@dist: ~/data   hduser@dist: ~/data
res12: Array[(Flight, FlightMetadata)] = Array((Flight(United States,Uganda,1),FlightMetadata(1,2764182393313317269)), (
Flight(United States,French Guiana,1),FlightMetadata(1,2764182393313317269)))

scala> val flights2 = flights.join(flightsMeta, Seq("count"))
flights2: org.apache.spark.sql.DataFrame = [count: bigint, DEST_COUNTRY_NAME: string ... 2 more fields]

scala> flights2.take(2)
res13: Array[org.apache.spark.sql.Row] = Array([1,United States,Uganda,5767147658854483195], [1,United States,French Gui
ana,5767147658854483195])

scala> flights.groupBy("DEST_COUNTRY_NAME").count()
res14: org.apache.spark.sql.DataFrame = [DEST_COUNTRY_NAME: string, count: bigint]

scala> flights.groupByKey(x => x.DEST_COUNTRY_NAME).count()
res15: org.apache.spark.sql.Dataset[(String, Long)] = [key: string, count(1): bigint]

scala> def sum2 (left:Flight, right:Flight) = {
     |   Flight(left.DEST_COUNTRY_NAME, null, left.count + right.count)
     | }
sum2: (left: Flight, right: Flight)Flight

scala> flights.groupByKey(x => x.DEST_COUNTRY_NAME).reduceGroups((l,r) => sum2(l,r)).take(5)
res16: Array[(String, Flight)] = Array((Anguilla,Flight(Anguilla,United States,21)), (Russia,Flight(Russia,United States
,152)), (Paraguay,Flight(Paraguay,United States,90)), (Senegal,Flight(Senegal,United States,29)), (Sweden,Flight(Sweden,
United States,65)))

scala>
```

```
Windows PowerShell   hduser@dist: ~/data   hduser@dist: ~/data

scala> flights.groupBy("DEST_COUNTRY_NAME").count().explain
== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- HashAggregate(keys=[DEST_COUNTRY_NAME#617], functions=[count(1)])
   +- Exchange hashpartitioning(DEST_COUNTRY_NAME#617, 200), ENSURE_REQUIREMENTS, [plan_id=694]
      +- HashAggregate(keys=[DEST_COUNTRY_NAME#617], functions=[partial_count(1)])
         +- FileScan parquet [DEST_COUNTRY_NAME#617] Batched: true, DataFilters: [], Format: Parquet, Location: InMemory
FileIndex(1 paths)[file:/home/hduser/data/part-r-00000.gz.parquet], PartitionFilters: [], PushedFilters: [], ReadSchema:
 struct<DEST_COUNTRY_NAME:string>



scala> flights.groupByKey(x => x.DEST_COUNTRY_NAME).count().explain
== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- HashAggregate(keys=[value#778], functions=[count(1)])
   +- Exchange hashpartitioning(value#778, 200), ENSURE_REQUIREMENTS, [plan_id=715]
      +- HashAggregate(keys=[value#778], functions=[partial_count(1)])
         +- Project [value#778]
            +- AppendColumns $line50.$read$$iw$$iw$$iw$$iw$$iw$$iw$$iw$$iw$$Lambda$5490/240035216@142fad99, newInstance(
class $line24.$read$$iw$$iw$$iw$$iw$$iw$$iw$$iw$$iw$Flight), [staticinvoke(class org.apache.spark.unsafe.types.UTF8Strin
g, StringType, fromString, input[0, java.lang.String, true], true, false, true) AS value#778]
               +- FileScan parquet [DEST_COUNTRY_NAME#617,ORIGIN_COUNTRY_NAME#618,count#619L] Batched: true, DataFilters
: [], Format: Parquet, Location: InMemoryFileIndex(1 paths)[file:/home/hduser/data/part-r-00000.gz.parquet], PartitionFi
lters: [], PushedFilters: [], ReadSchema: struct<DEST_COUNTRY_NAME:string,ORIGIN_COUNTRY_NAME:string,count:bigint>
```

```
Windows PowerShell   hduser@dist: ~/data
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 1.8.0_452)
Type in expressions to have them evaluated.
Type :help for more information.

scala> val df = spark.read.format("json").load("./2015-summary.json")
df: org.apache.spark.sql.DataFrame = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ... 1 more field]

scala> df.select(count("DEST_COUNTRY_NAME"), countDistinct("DEST_COUNTRY_NAME")).show()
+-----------------------+---------------------------------+
|count(DEST_COUNTRY_NAME)|count(DISTINCT DEST_COUNTRY_NAME)|
+-----------------------+---------------------------------+
|                    256|                              132|
+-----------------------+---------------------------------+


scala> df.select(min("count"), max("count"), sum("count"), sumDistinct("count"),avg("count")).show()
warning: one deprecation (since 3.2.0); for details, enable `:setting -deprecation' or `:replay -deprecation'
+----------+----------+----------+-------------------+-----------+
|min(count)|max(count)|sum(count)|sum(DISTINCT count)| avg(count)|
+----------+----------+----------+-------------------+-----------+
|         1|    370002|    453316|             450718|1770.765625|
+----------+----------+----------+-------------------+-----------+


scala>
```

```
scala> df.select(count("DEST_COUNTRY_NAME"), countDistinct("DEST_COUNTRY_NAME")).show()
+-----------------------+-------------------------------+
|count(DEST_COUNTRY_NAME)|count(DISTINCT DEST_COUNTRY_NAME)|
+-----------------------+-------------------------------+
|                    256|                            132|
+-----------------------+-------------------------------+


scala> df.select(min("count"), max("count"), sum("count"), sumDistinct("count"),avg("count")).show()
warning: one deprecation (since 3.2.0); for details, enable `:setting -deprecation' or `:replay -deprecation'
+----------+----------+----------+------------------+-----------+
|min(count)|max(count)|sum(count)|sum(DISTINCT count)| avg(count)|
+----------+----------+----------+------------------+-----------+
|         1|    370002|    453316|            450718|1770.765625|
+----------+----------+----------+------------------+-----------+


scala> df.select(var_pop("count"),var_samp("count"),stddev_pop("count"),stddev_samp("count")).show()
+--------------------+--------------------+-----------------+------------------+
|      var_pop(count)|    var_samp(count)|stddev_pop(count)|stddev_samp(count)|
+--------------------+--------------------+-----------------+------------------+
|5.3274658249975514E8|5.348357847840679E8|23081.30374350104|23126.516918551915|
+--------------------+--------------------+-----------------+------------------+


scala>
```

```
scala> df.groupBy("DEST_COUNTRY_NAME").agg(sum("count").alias("sum_count")).show()
+--------------------+----------+
|   DEST_COUNTRY_NAME|sum_count|
+--------------------+----------+
|            Anguilla|        41|
|             Russia|       176|
|           Paraguay|        60|
|            Senegal|        40|
|             Sweden|       118|
|           Kiribati|        26|
|             Guyana|        64|
|        Philippines|       134|
|           Djibouti|         1|
|           Malaysia|         2|
|          Singapore|         3|
|               Fiji|        24|
|             Turkey|       138|
|               Iraq|         1|
|            Germany|      1468|
|             Jordan|        44|
|              Palau|        30|
|Turks and Caicos ...|       230|
|             France|       935|
|             Greece|        30|
+--------------------+----------+
only showing top 20 rows
```

```
scala> df.groupBy("DEST_COUNTRY_NAME").agg(sum("count").alias("sum_count")).orderBy(col("sum_count").desc).show()
+-------------------+----------+
| DEST_COUNTRY_NAME|sum_count|
+-------------------+----------+
|     United States|    411352|
|            Canada|      8399|
|            Mexico|      7140|
|    United Kingdom|      2025|
|             Japan|      1548|
|           Germany|      1468|
|Dominican Republic|      1353|
|       South Korea|      1048|
|       The Bahamas|       955|
|            France|       935|
|          Colombia|       873|
|            Brazil|       853|
|       Netherlands|       776|
|             China|       772|
|           Jamaica|       666|
|        Costa Rica|       588|
|       El Salvador|       561|
|            Panama|       510|
|              Cuba|       466|
|             Spain|       420|
+-------------------+----------+
only showing top 20 rows
```

```
scala> df.groupBy("DEST_COUNTRY_NAME").agg(sum("count").alias("sum_count"),avg("count").alias("avg_count")).orderBy(col(
"sum_count").desc).show()
+-------------------+---------+---------+
| DEST_COUNTRY_NAME|sum_count|avg_count|
+-------------------+---------+---------+
|     United States|   411352| 3290.816|
|            Canada|     8399|   8399.0|
|            Mexico|     7140|   7140.0|
|    United Kingdom|     2025|   2025.0|
|             Japan|     1548|   1548.0|
|           Germany|     1468|   1468.0|
|Dominican Republic|     1353|   1353.0|
|       South Korea|     1048|   1048.0|
|       The Bahamas|      955|    955.0|
|            France|      935|    935.0|
|          Colombia|      873|    873.0|
|            Brazil|      853|    853.0|
|       Netherlands|      776|    776.0|
|             China|      772|    772.0|
|           Jamaica|      666|    666.0|
|        Costa Rica|      588|    588.0|
|       El Salvador|      561|    561.0|
|            Panama|      510|    510.0|
|              Cuba|      466|    466.0|
|             Spain|      420|    420.0|
+-------------------+---------+---------+
only showing top 20 rows
```

```
Time taken: 3.707 seconds
spark-sql (default)> create table flight1 (
                   >   DEST_COUNTRY_NAME STRING,
                   >   ORIGIN_COUNTRY_NAME STRING,
                   >   count LONG
                   > ) using json options (
                   >   path '../../../2015-summary.json'
                   > );
[TABLE_OR_VIEW_ALREADY_EXISTS] Cannot create table or view `spark_catalog`.`default`.`flight1` because it already exists
.
Choose a different name, drop or replace the existing object, or add the IF NOT EXISTS clause to tolerate pre-existing o
bjects.
spark-sql (default)> drop table flight1;
25/06/04 14:32:13 WARN ObjectStore: Failed to get database global_temp, returning NoSuchObjectException
Time taken: 0.952 seconds
spark-sql (default)> create table flight1 (
                   >   DEST_COUNTRY_NAME STRING,
                   >   ORIGIN_COUNTRY_NAME STRING,
                   >   count LONG
                   > ) using json options (
                   >   path '../../../2015-summary.json'
                   > );
25/06/04 14:32:16 WARN HiveExternalCatalog: Couldn't find corresponding Hive SerDe for data source provider json. Persis
ting data source table `spark_catalog`.`default`.`flight1` into Hive metastore in Spark SQL specific format, which is NO
T compatible with Hive.
Time taken: 0.243 seconds
spark-sql (default)>
```

```
Time taken: 4.392 seconds
spark-sql (default)>
                   > ;
spark-sql (default)> describe table extended flight2;
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint

# Detailed Table Information
Catalog                 spark_catalog
Database                default
Table                   flight2
Owner                   hduser
Created Time            Wed Jun 04 14:33:36 UTC 2025
Last Access             UNKNOWN
Created By              Spark 3.5.5
Type                    EXTERNAL
Provider                hive
Table Properties        [transient_lastDdlTime=1749047619]
Location                file:/home/hduser/data/spark/sql/flight2
Serde Library           org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe
InputFormat             org.apache.hadoop.mapred.TextInputFormat
OutputFormat            org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat
Storage Properties      [serialization.format=,, field.delim=,]
Partition Provider      Catalog
Time taken: 0.3 seconds, Fetched 21 row(s)
spark-sql (default)>
```

```
OutputFormat            org.apache.hadoop.hive.ql.io.HiveSequenceFileOutputFormat
Storage Properties      [serialization.format=1]
Time taken: 0.147 seconds, Fetched 22 row(s)
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint

# Detailed Table Information
Catalog                 spark_catalog
Database                default
Table                   usa_flight
Owner                   hduser
Created Time            Wed Jun 04 14:34:10 UTC 2025
Last Access             UNKNOWN
Created By              Spark 3.5.5
Type                    VIEW
View Text               select * from flight where dest_country_name='United States'
View Original Text      select * from flight where dest_country_name='United States'
View Catalog and Namespace      spark_catalog.default
View Query Output Columns       [DEST_COUNTRY_NAME, ORIGIN_COUNTRY_NAME, count]
Table Properties        [transient_lastDdlTime=1749047650]
Serde Library           org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe
InputFormat             org.apache.hadoop.mapred.SequenceFileInputFormat
OutputFormat            org.apache.hadoop.hive.ql.io.HiveSequenceFileOutputFormat
Storage Properties      [serialization.format=1]
Time taken: 0.126 seconds, Fetched 22 row(s)
spark-sql (default)>
```

```
25/06/04 14:37:22 WARN ObjectStore: Version information not found in metastore. hive.metastore.schema.verification is no
t enabled so recording the schema version 2.3.0
25/06/04 14:37:22 WARN ObjectStore: setMetaStoreSchemaVersion called but recording version is disabled: version = 2.3.0,
 comment = Set by MetaStore hduser@10.0.2.15
res0: org.apache.spark.sql.DataFrame = [namespace: string]

scala> spark.sql("show databases").show()
+---------+
|namespace|
+---------+
|  default|
|  testdb1|
+---------+


scala> spark.sql("use testdb1")
25/06/04 14:37:44 WARN ObjectStore: Failed to get database global_temp, returning NoSuchObjectException
res2: org.apache.spark.sql.DataFrame = []

scala> spark.sql("show tables").show()
+---------+---------+-----------+
|namespace|tableName|isTemporary|
+---------+---------+-----------+
+---------+---------+-----------+


scala>
```

```
|   DEST_COUNTRY_NAME|ORIGIN_COUNTRY_NAME|count|
+--------------------+-------------------+-----+
|       United States|            Romania|   15|
|       United States|            Croatia|    1|
|       United States|            Ireland|  344|
|              Egypt|      United States|   15|
|       United States|              India|   62|
|       United States|          Singapore|    1|
|       United States|            Grenada|   62|
|          Costa Rica|      United States|  588|
|             Senegal|      United States|   40|
|             Moldova|      United States|    1|
|       United States|       Sint Maarten|  325|
|       United States|   Marshall Islands|   39|
|              Guyana|      United States|   64|
|               Malta|      United States|    1|
|            Anguilla|      United States|   41|
|             Bolivia|      United States|   30|
|       United States|           Paraguay|    6|
|             Algeria|      United States|    4|
|Turks and Caicos ...|      United States|  230|
|       United States|          Gibraltar|    1|
+--------------------+-------------------+-----+
only showing top 20 rows


scala>
```

```
25/06/04 14:47:46 WARN HiveConf: HiveConf of name hive.stats.max.num.stats does not exist
25/06/04 14:47:46 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
df: org.apache.spark.sql.DataFrame = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ... 1 more field]

scala> df.show();
+-------------------+-------------------+-----+
|   DEST_COUNTRY_NAME|ORIGIN_COUNTRY_NAME|count|
+-------------------+-------------------+-----+
|       United States|            Romania|   15|
|       United States|            Croatia|    1|
|       United States|            Ireland|  344|
|              Egypt|      United States|   15|
|       United States|              India|   62|
|       United States|          Singapore|    1|
|       United States|            Grenada|   62|
|          Costa Rica|      United States|  588|
|            Senegal|      United States|   40|
|            Moldova|      United States|    1|
|       United States|       Sint Maarten|  325|
|       United States|   Marshall Islands|   39|
|             Guyana|      United States|   64|
|              Malta|      United States|    1|
|           Anguilla|      United States|   41|
|            Bolivia|      United States|   30|
|       United States|           Paraguay|    6|
|            Algeria|      United States|    4|
|Turks and Caicos ...|      United States|  230|
```

```
scala> df.first();
res2: org.apache.spark.sql.Row = [United States,Romania,15]

scala>

scala> df.collect();
25/06/04 14:48:11 WARN LazyStruct: Extra bytes detected at the end of the row! Ignoring similar problems.
res3: Array[org.apache.spark.sql.Row] = Array([United States,Romania,15], [United States,Croatia,1], [United States,Irel
and,344], [Egypt,United States,15], [United States,India,62], [United States,Singapore,1], [United States,Grenada,62], [
Costa Rica,United States,588], [Senegal,United States,40], [Moldova,United States,1], [United States,Sint Maarten,325],
[United States,Marshall Islands,39], [Guyana,United States,64], [Malta,United States,1], [Anguilla,United States,41], [B
olivia,United States,30], [United States,Paraguay,6], [Algeria,United States,4], [Turks and Caicos Islands,United States
,230], [United States,Gibraltar,1], [Saint Vincent and the Grenadines,United States,1], [Italy,United States,382], [Unit
ed States,Federated States of Micronesia,69], [Unit...

scala> df.filter(col("count") > 10).show(2);
+-----------------+-------------------+-----+
|DEST_COUNTRY_NAME|ORIGIN_COUNTRY_NAME|count|
+-----------------+-------------------+-----+
|    United States|            Romania|   15|
|    United States|            Ireland|  344|
+-----------------+-------------------+-----+
only showing top 2 rows


scala>
```

```
scala> df.groupBy("DEST_COUNTRY_NAME").agg(
     |   sum("count").alias("sum_count"),
     |   avg("count").alias("avg_count")
     | ).orderBy(col("sum_count").desc).show();
25/06/04 14:50:19 WARN LazyStruct: Extra bytes detected at the end of the row! Ignoring similar problems.
+-------------------+---------+-------------------+
| DEST_COUNTRY_NAME|sum_count|          avg_count|
+-------------------+---------+-------------------+
|     United States|   411293|3316.8790322580644|
|            Canada|     8399|             8399.0|
|            Mexico|     7140|             7140.0|
|    United Kingdom|     2025|             2025.0|
|             Japan|     1548|             1548.0|
|           Germany|     1468|             1468.0|
|Dominican Republic|     1353|             1353.0|
|       South Korea|     1048|             1048.0|
|       The Bahamas|      955|              955.0|
|            France|      935|              935.0|
|          Colombia|      873|              873.0|
|            Brazil|      853|              853.0|
|       Netherlands|      776|              776.0|
|             China|      772|              772.0|
|           Jamaica|      666|              666.0|
|        Costa Rica|      588|              588.0|
|       El Salvador|      561|              561.0|
|            Panama|      510|              510.0|
|              Cuba|      466|              466.0|
```

```
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.use.orc.codec.pool does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.query.results.cache.max.size does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.auto.convert.anti.join does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.strict.timestamp.conversion does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.repl.bootstrap.dump.open.txn.timeout does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.cardinality.preserving.join.optimization.factor does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.stats.max.num.stats does not exist
25/06/04 14:23:50 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
25/06/04 14:23:54 WARN ObjectStore: Version information not found in metastore. hive.metastore.schema.verification is no
t enabled so recording the schema version 2.3.0
25/06/04 14:23:54 WARN ObjectStore: setMetaStoreSchemaVersion called but recording version is disabled: version = 2.3.0,
 comment = Set by MetaStore hduser@10.0.2.15
25/06/04 14:23:54 WARN ObjectStore: Failed to get database default, returning NoSuchObjectException
Spark Web UI available at http://dist:4040
Spark master: local[*], Application Id: local-1749047023060
spark-sql (default)> show databases;
default
Time taken: 2.9 seconds, Fetched 1 row(s)
spark-sql (default)> create database testdb1;
25/06/04 14:24:21 WARN ObjectStore: Failed to get database testdb1, returning NoSuchObjectException
25/06/04 14:24:21 WARN ObjectStore: Failed to get database testdb1, returning NoSuchObjectException
25/06/04 14:24:21 WARN ObjectStore: Failed to get database global_temp, returning NoSuchObjectException
25/06/04 14:24:21 WARN ObjectStore: Failed to get database testdb1, returning NoSuchObjectException
Time taken: 1.351 seconds
spark-sql (default)> use testdb1;
Time taken: 0.091 seconds
spark-sql (testdb1)>
```

```
25/06/04 14:24:43 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
25/06/04 14:24:43 WARN HiveMetaStore: Location: file:/home/hduser/data/spark/sql/spark-warehouse/testdb1.db/flight speci
fied for non-external table:flight
Time taken: 1.349 seconds
spark-sql (testdb1)> show tables;
flight
Time taken: 0.249 seconds, Fetched 1 row(s)
spark-sql (testdb1)> describe table flight;
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint
Time taken: 0.49 seconds, Fetched 3 row(s)
spark-sql (testdb1)> describe table extended flight;
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint

# Detailed Table Information
Catalog                 spark_catalog
Database                testdb1
Table                   flight
Owner                   hduser
Created Time            Wed Jun 04 14:24:43 UTC 2025
Last Access             UNKNOWN
Created By              Spark 3.5.5
Type                    MANAGED
Provider                hive
```

```
Time taken: 0.49 seconds, Fetched 3 row(s)
spark-sql (testdb1)> describe table extended flight;
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint

# Detailed Table Information
Catalog                 spark_catalog
Database                testdb1
Table                   flight
Owner                   hduser
Created Time            Wed Jun 04 14:24:43 UTC 2025
Last Access             UNKNOWN
Created By              Spark 3.5.5
Type                    MANAGED
Provider                hive
Table Properties        [transient_lastDdlTime=1749047083]
Location                file:/home/hduser/data/spark/sql/spark-warehouse/testdb1.db/flight
Serde Library           org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe
InputFormat             org.apache.hadoop.mapred.TextInputFormat
OutputFormat            org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat
Storage Properties      [serialization.format=1]
Partition Provider      Catalog
Time taken: 0.18 seconds, Fetched 21 row(s)
spark-sql (testdb1)> drop table flight;
Time taken: 1.747 seconds
spark-sql (testdb1)>
```

```
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.server2.show.operation.drilldown.link does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.txn.acid.metrics.reporting.interval does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.zookeeper.killquery.namespace does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.repl.copy.file.list.iterator.retry does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.optimize.topnkey.efficiency.threshold does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.server2.webui.explain.output does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.llap.plugin.rpc.port does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.lock.query.string.max.length does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.llap.io.track.cache.usage does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.server2.authentication.jwt.jwks.skip.ssl.cert does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.repl.retry.initial.delay does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.hook.proto.events.clean.freq does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.use.orc.codec.pool does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.query.results.cache.max.size does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.auto.convert.anti.join does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.strict.timestamp.conversion does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.repl.bootstrap.dump.open.txn.timeout does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.cardinality.preserving.join.optimization.factor does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.stats.max.num.stats does not exist
25/06/04 14:42:07 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
25/06/04 14:42:07 WARN HiveMetaStore: Location: file:/home/hduser/data/spark/sql/spark-warehouse/flight specified for no
n-external table:flight
Time taken: 0.646 seconds
spark-sql (default)>
                    > load data local inpath '/home/hduser/data/test.csv' into table flight;
Time taken: 0.47 seconds
spark-sql (default)>
```

```
United States   Bermuda 193
Chile   United States   174
United States   Cuba    478
United States   Montenegro      1
United States   Colombia        867
United States   Barbados        130
United States   Qatar   109
Australia       United States   329
United States   Cayman Islands  310
United States   Jordan  44
United States   Namibia 1
United States   Trinidad and Tobago     217
United States   Bolivia 13
Cook Islands    United States   13
Bulgaria        United States   3
United States   Saint Kitts and Nevis   145
Uruguay United States   43
United States   Haiti   225
"Bonaire         Sint Eustatius NULL
Greece  United States   30
Time taken: 2.831 seconds, Fetched 256 row(s)
spark-sql (default)> describe table flight;
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint
Time taken: 0.212 seconds, Fetched 3 row(s)
spark-sql (default)> describe table extended flight;
```

```
count                   bigint
Time taken: 0.212 seconds, Fetched 3 row(s)
spark-sql (default)> describe table extended flight;
DEST_COUNTRY_NAME       string
ORIGIN_COUNTRY_NAME     string
count                   bigint

# Detailed Table Information
Catalog                 spark_catalog
Database                default
Table                   flight
Owner                   hduser
Created Time            Wed Jun 04 14:42:07 UTC 2025
Last Access             UNKNOWN
Created By              Spark 3.5.5
Type                    MANAGED
Provider                hive
Table Properties        [transient_lastDdlTime=1749048128]
Statistics              7036 bytes
Location                file:/home/hduser/data/spark/sql/spark-warehouse/flight
Serde Library           org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe
InputFormat             org.apache.hadoop.mapred.TextInputFormat
OutputFormat            org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat
Storage Properties      [serialization.format=,, field.delim=,]
Partition Provider      Catalog
Time taken: 0.332 seconds, Fetched 22 row(s)
spark-sql (default)>
```

```
Windows PowerShell   hduser@dist: ~/data/streamin   hduser@dist: ~/data/spark/sq
25/06/04 14:56:58 INFO Executor: Finished task 2.0 in stage 15.0 (TID 15). 1705 bytes result sent to driver
25/06/04 14:56:58 INFO Executor: Finished task 1.0 in stage 15.0 (TID 14). 1748 bytes result sent to driver
25/06/04 14:56:58 INFO TaskSetManager: Finished task 2.0 in stage 15.0 (TID 15) in 73 ms on dist (executor driver) (2/3)
25/06/04 14:56:58 INFO TaskSetManager: Finished task 1.0 in stage 15.0 (TID 14) in 88 ms on dist (executor driver) (3/3)
25/06/04 14:56:58 INFO TaskSchedulerImpl: Removed TaskSet 15.0, whose tasks have all completed, from pool
25/06/04 14:56:58 INFO DAGScheduler: ResultStage 15 (print at HdfsWordCount.scala:52) finished in 0.153 s
25/06/04 14:56:58 INFO DAGScheduler: Job 7 is finished. Cancelling potential speculative or zombie tasks for this job
25/06/04 14:56:58 INFO TaskSchedulerImpl: Killing all running tasks in stage 15: Stage finished
25/06/04 14:56:58 INFO DAGScheduler: Job 7 finished: print at HdfsWordCount.scala:52, took 0.176748 s
-------------------------------------------
Time: 1749049018000 ms
-------------------------------------------

25/06/04 14:56:58 INFO JobScheduler: Finished job streaming job 1749049018000 ms.0 from job set of time 1749049018000 ms
25/06/04 14:56:58 INFO JobScheduler: Total delay: 0.368 s for time 1749049018000 ms (execution: 0.321 s)
25/06/04 14:56:58 INFO ShuffledRDD: Removing RDD 14 from persistence list
25/06/04 14:56:58 INFO MapPartitionsRDD: Removing RDD 13 from persistence list
25/06/04 14:56:58 INFO BlockManager: Removing RDD 14
25/06/04 14:56:58 INFO MapPartitionsRDD: Removing RDD 12 from persistence list
25/06/04 14:56:58 INFO BlockManager: Removing RDD 13
25/06/04 14:56:58 INFO MapPartitionsRDD: Removing RDD 11 from persistence list
25/06/04 14:56:58 INFO FileInputDStream: Cleared 0 old files that were older than 1749048958000 ms:
25/06/04 14:56:58 INFO ReceivedBlockTracker: Deleting batches:
25/06/04 14:56:58 INFO BlockManager: Removing RDD 12
25/06/04 14:56:58 INFO InputInfoTracker: remove old batch metadata:
25/06/04 14:56:58 INFO BlockManager: Removing RDD 11
```

```
Windows PowerShell   hduser@dist: ~/data/streamin   hduser@dist: ~/data/streamin
Table                   flight
Owner                   hduser
Created Time            Wed Jun 04 14:42:07 UTC 2025
Last Access             UNKNOWN
Created By              Spark 3.5.5
Type                    MANAGED
Provider                hive
Table Properties        [transient_lastDdlTime=1749048128]
Statistics              7036 bytes
Location                file:/home/hduser/data/spark/sql/spark-warehouse/flight
Serde Library           org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe
InputFormat             org.apache.hadoop.mapred.TextInputFormat
OutputFormat            org.apache.hadoop.hive.ql.io.HiveIgnoreKeyTextOutputFormat
Storage Properties      [serialization.format=,, field.delim=,]
Partition Provider      Catalog
Time taken: 3.335 seconds, Fetched 22 row(s)
spark-sql (default)> hduser@dist:~/data/spark/sql$ ls
derby.log  flight2  metastore_db  spark-warehouse
hduser@dist:~/data/spark/sql$ cd ../../
hduser@dist:~/data$ ls
2015-summary.csv   derby.log  hbase         part-r-00000.gz.parquet  streaming_test
2015-summary.json  emp.csv    metastore_db  spark                    test.csv
hduser@dist:~/data$ cd streaming_test/
hduser@dist:~/data/streaming_test$ vi new-awesome
hduser@dist:~/data/streaming_test$ ls
new-awesome
hduser@dist:~/data/streaming_test$
```

```
Windows PowerShell   hduser@dist: ~/data/streamin   hduser@dist: ~/data/streamin
Count.scala:69), which has no missing parents
25/06/04 14:59:25 INFO MemoryStore: Block broadcast_2 stored as values in memory (estimated size 41.7 KiB, free 365.5 Mi
B)
25/06/04 14:59:25 INFO MemoryStore: Block broadcast_2_piece0 stored as bytes in memory (estimated size 18.5 KiB, free 36
5.5 MiB)
25/06/04 14:59:25 INFO BlockManagerInfo: Added broadcast_2_piece0 in memory on dist:39463 (size: 18.5 KiB, free: 366.2 M
iB)
25/06/04 14:59:25 INFO SparkContext: Created broadcast 2 from broadcast at DAGScheduler.scala:1585
25/06/04 14:59:25 INFO DAGScheduler: Submitting 4 missing tasks from ShuffleMapStage 0 (MapPartitionsRDD[6] at start at
StructuredNetworkWordCount.scala:69) (first 15 tasks are for partitions Vector(0, 1, 2, 3))
25/06/04 14:59:25 INFO TaskSchedulerImpl: Adding task set 0.0 with 4 tasks resource profile 0
25/06/04 14:59:25 INFO TaskSetManager: Starting task 0.0 in stage 0.0 (TID 0) (dist, executor driver, partition 0, PROCE
SS_LOCAL, 9615 bytes)
25/06/04 14:59:25 INFO TaskSetManager: Starting task 1.0 in stage 0.0 (TID 1) (dist, executor driver, partition 1, PROCE
SS_LOCAL, 9615 bytes)
25/06/04 14:59:25 INFO TaskSetManager: Starting task 2.0 in stage 0.0 (TID 2) (dist, executor driver, partition 2, PROCE
SS_LOCAL, 9615 bytes)
25/06/04 14:59:25 INFO TaskSetManager: Starting task 3.0 in stage 0.0 (TID 3) (dist, executor driver, partition 3, PROCE
SS_LOCAL, 9615 bytes)
25/06/04 14:59:25 INFO Executor: Running task 2.0 in stage 0.0 (TID 2)
25/06/04 14:59:25 INFO Executor: Running task 1.0 in stage 0.0 (TID 1)
25/06/04 14:59:25 INFO Executor: Running task 3.0 in stage 0.0 (TID 3)
25/06/04 14:59:25 INFO Executor: Running task 0.0 in stage 0.0 (TID 0)
25/06/04 14:59:26 INFO CodeGenerator: Code generated in 22.085339 ms
25/06/04 14:59:26 INFO CodeGenerator: Code generated in 17.15629 ms
25/06/04 14:59:26 INFO CodeGenerator: Code generated in 64.675756 ms
```

```
12-def1e58da7c2.tmp
25/06/04 14:59:44 INFO CheckpointFileManager: Renamed temp file file:/tmp/temporary-280a879f-46a2-4611-a518-95f23955dd50
/commits/.0.8bb0f3fd-57cf-4b48-8e12-def1e58da7c2.tmp to file:/tmp/temporary-280a879f-46a2-4611-a518-95f23955dd50/commits
/0
25/06/04 14:59:44 INFO MicroBatchExecution: Streaming query made progress: {
  "id" : "3bcfd800-ff53-4be5-ac26-43c9a4b39af8",
  "runId" : "9ab827e2-0a79-4f7e-ae06-05f1e364fd51",
  "name" : null,
  "timestamp" : "2025-06-04T14:59:20.529Z",
  "batchId" : 0,
  "numInputRows" : 0,
  "inputRowsPerSecond" : 0.0,
  "processedRowsPerSecond" : 0.0,
  "durationMs" : {
    "addBatch" : 22457,
    "commitOffsets" : 103,
    "getBatch" : 2,
    "latestOffset" : 0,
    "queryPlanning" : 1329,
    "triggerExecution" : 24119,
    "walCommit" : 180
  },
  "stateOperators" : [ {
    "operatorName" : "stateStoreSave",
    "numRowsTotal" : 0,
    "numRowsUpdated" : 0,
    "allUpdatesTimeMs" : 2201,
```

```
    "allRemovalsTimeMs" : 0,
    "commitTimeMs" : 41036,
    "memoryUsedBytes" : 82336,
    "numRowsDroppedByWatermark" : 0,
    "numShufflePartitions" : 200,
    "numStateStoreInstances" : 200,
    "customMetrics" : {
      "loadedMapCacheHitCount" : 400,
      "loadedMapCacheMissCount" : 0,
      "stateOnCurrentVersionSizeBytes" : 21536
    }
  } ],
  "sources" : [ {
    "description" : "TextSocketV2[host: localhost, port: 9999]",
    "startOffset" : -1,
    "endOffset" : 4,
    "latestOffset" : 4,
    "numInputRows" : 5,
    "inputRowsPerSecond" : 0.20638982910922152,
    "processedRowsPerSecond" : 0.3152386356471849
  } ],
  "sink" : {
    "description" : "org.apache.spark.sql.execution.streaming.ConsoleTable$@350ec63",
    "numOutputRows" : 4
  }
}
```

```
 * and then run the example
 *    `$ bin/run-example org.apache.spark.examples.streaming.NetworkWordCount localhost 9999`
 */
object NetworkWordCount {
  def main(args: Array[String]): Unit = {
    if (args.length < 2) {
      System.err.println("Usage: NetworkWordCount <hostname> <port>")
      System.exit(1)
    }

    StreamingExamples.setStreamingLogLevels()

    // Create the context with a 1 second batch size
    val sparkConf = new SparkConf().setAppName("NetworkWordCount")
    val ssc = new StreamingContext(sparkConf, Seconds(1))

    // Create a socket stream on target ip:port and count the
    // words in input stream of \n delimited text (e.g. generated by 'nc')
    // Note that no duplication in storage level only for running locally.
    // Replication necessary in distributed scenario for fault tolerance.
    val lines = ssc.socketTextStream(args(0), args(1).toInt, StorageLevel.MEMORY_AND_DISK_SER)
    val words = lines.flatMap(_.split(" "))
    val wordCounts = words.map(x => (x, 1)).reduceByKey(_ + _)
    wordCounts.print()
    ssc.start()
    ssc.awaitTermination()
/val sparkConf
```

```
Windows PowerShell   hduser@dist: ~/data/spark/sq   hduser@dist: ~/data/spark/sq
25/06/04 14:52:28 INFO DAGScheduler: ResultStage 162 (print at NetworkWordCount.scala:56) failed in 2.357 s due to Stage
 cancelled because SparkContext was shut down
25/06/04 14:52:28 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
25/06/04 14:52:28 INFO MemoryStore: MemoryStore cleared
25/06/04 14:52:28 INFO BlockManager: BlockManager stopped
25/06/04 14:52:28 INFO BlockManagerMaster: BlockManagerMaster stopped
25/06/04 14:52:28 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
25/06/04 14:52:28 INFO SparkContext: Successfully stopped SparkContext
25/06/04 14:52:28 INFO ShutdownHookManager: Shutdown hook called
25/06/04 14:52:28 INFO ShutdownHookManager: Deleting directory /tmp/spark-abf0aa84-88e3-4ece-876d-c897460415ea
25/06/04 14:52:28 INFO ShutdownHookManager: Deleting directory /tmp/spark-656ff337-f2f8-44e9-b607-aae9eb85b45c
hduser@dist:~/data/spark/sql$ $SPARK_HOME/examples/src/main/scala/org/apache/spark/examples/streaming
-bash: /home/hduser/spark-3.5.5-bin-hadoop3/examples/src/main/scala/org/apache/spark/examples/streaming: Is a directory
hduser@dist:~/data/spark/sql$ ls $SPARK_HOME/examples/src/main/scala/org/apache/spark/examples/streaming
clickstream                       HdfsWordCount.scala     RecoverableNetworkWordCount.scala
CustomReceiver.scala              NetworkWordCount.scala  SqlNetworkWordCount.scala
DirectKafkaWordCount.scala        QueueStream.scala       StatefulNetworkWordCount.scala
DirectKerberizedKafkaWordCount.scala  RawNetworkGrep.scala    StreamingExamples.scala
hduser@dist:~/data/spark/sql$ cat $SPARK_HOME/examples/src/main/scala/org/apache/spark/examples/streaming/NetworkWordCou
nt.scala
/*
 * Licensed to the Apache Software Foundation (ASF) under one or more
 * contributor license agreements.  See the NOTICE file distributed with
 * this work for additional information regarding copyright ownership.
 * The ASF licenses this file to You under the Apache License, Version 2.0
 * (the "License"); you may not use this file except in compliance with
 * the License.  You may obtain a copy of the License at
```

```
azureuser@test03: ~/hadoop-3.4.1/etc/hadoop
doop-3.4.1/namenode/current/VERSION]
2025-06-06 17:03:06,045 INFO common.Storage: Storage directory /home/azureuser/h
adoop-3.4.1/namenode has been successfully formatted.
2025-06-06 17:03:06,136 INFO namenode.FSImageFormatProtobuf: Saving image file /
home/azureuser/hadoop-3.4.1/namenode/current/fsimage.ckpt_0000000000000000000 us
ing no compression
2025-06-06 17:03:06,345 INFO namenode.FSImageFormatProtobuf: Image file /home/az
ureuser/hadoop-3.4.1/namenode/current/fsimage.ckpt_0000000000000000000 of size 4
01 bytes saved in 0 seconds .
2025-06-06 17:03:06,364 INFO namenode.NNStorageRetentionManager: Going to retain
 1 images with txid >= 0
2025-06-06 17:03:06,376 INFO blockmanagement.DatanodeManager: Slow peers collect
ion thread shutdown
2025-06-06 17:03:06,386 INFO namenode.FSNamesystem: Stopping services started fo
r active state
2025-06-06 17:03:06,386 INFO namenode.FSNamesystem: Stopping services started fo
r standby state
2025-06-06 17:03:06,395 INFO namenode.FSImage: FSImageSaver clean checkpoint: tx
id=0 when meet shutdown.
2025-06-06 17:03:06,396 INFO namenode.NameNode: SHUTDOWN_MSG:
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at test03/20.0.121.104
************************************************************/
azureuser@test03:~/hadoop-3.4.1/etc/hadoop$
```

```
azureuser@test03: ~
2025-06-06 17:10:10,269 INFO namenode.FSImage: FSImageSaver clean checkpoint: tx
id=0 when meet shutdown.
2025-06-06 17:10:10,270 INFO namenode.NameNode: SHUTDOWN_MSG:
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at test03/20.0.121.104
************************************************************/
azureuser@test03:~$ start-dfs.sg]
start-dfs.sg]: command not found
azureuser@test03:~$ ^C
azureuser@test03:~$ start-dfs.sh
Starting namenodes on [test03]
Starting datanodes
test04: datanode is running as process 5773.  Stop it first and ensure /tmp/hado
op-azureuser-datanode.pid file is empty before retry.
Starting secondary namenodes [test03]
azureuser@test03:~$ stop-dfs.sh
Stopping namenodes on [test03]
Stopping datanodes
Stopping secondary namenodes [test03]
azureuser@test03:~$ start-dfs.sh
Starting namenodes on [test03]
Starting datanodes
Starting secondary namenodes [test03]
azureuser@test03:~$
```

```
Windows PowerShell   hduser@dist: /media/sf_dist   hduser@dist: ~
        at org.apache.spark.deploy.SparkSubmit.doSubmit(SparkSubmit.scala:86)
        at org.apache.spark.deploy.SparkSubmit$$anon$2.doSubmit(SparkSubmit.scala:1125)
        at org.apache.spark.deploy.SparkSubmit$.main(SparkSubmit.scala:1134)
        at org.apache.spark.deploy.SparkSubmit.main(SparkSubmit.scala)
hduser@dist:~$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 2602.  Stop it first and ensure /tmp/hadoop-hduser-resourcemanager.pid file is emp
ty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 2729.  Stop it first and ensure /tmp/hadoop-hduser-nodemanager.pid file is
empty before retry.
hduser@dist:~$ source ~/.bashrc
hduser@dist:~$ spark-submit --master yarn --class org.apache.spark.examples.SparkPi --num-executors 3 --deploy-mode clie
nt $SPARK_HOME/examples/jars/sparkexamples_2.12-3.4.4.jar 100
25/06/06 18:02:10 WARN DependencyUtils: Local jar /home/hduser/spark-3.5.5-bin-hadoop3/examples/jars/sparkexamples_2.12-
3.4.4.jar does not exist, skipping.
Error: Failed to load class org.apache.spark.examples.SparkPi.
25/06/06 18:02:10 INFO ShutdownHookManager: Shutdown hook called
25/06/06 18:02:10 INFO ShutdownHookManager: Deleting directory /tmp/spark-9752408f-2303-4725-a61a-c3ea11b246ca
hduser@dist:~$ spark-submit --master yarn --class org.apache.spark.examples.SparkPi --deploy-mode client $SPARK_HOME/exa
mples/jars/sparkexamples_2.12-3.4.4.jar 100
25/06/06 18:03:00 WARN DependencyUtils: Local jar /home/hduser/spark-3.5.5-bin-hadoop3/examples/jars/sparkexamples_2.12-
3.4.4.jar does not exist, skipping.
Error: Failed to load class org.apache.spark.examples.SparkPi.
25/06/06 18:03:00 INFO ShutdownHookManager: Shutdown hook called
25/06/06 18:03:00 INFO ShutdownHookManager: Deleting directory /tmp/spark-7e9198c7-52b3-4b35-ad96-7b220233a834
hduser@dist:~$
```

```
Windows PowerShell   hduser@dist: /media/sf_dist   hduser@dist: ~/spark-3.5.5-bi
scopt_2.12-3.7.1.jar  spark-examples_2.12-3.5.5.jar
hduser@dist:~/spark-3.5.5-bin-hadoop3/examples/jars$
hduser@dist:~/spark-3.5.5-bin-hadoop3/examples/jars$
hduser@dist:~/spark-3.5.5-bin-hadoop3/examples/jars$ spark-submit --master yarn --class org.apache.spark.examples.SparkP
i --deploy-mode client $SPARK_HOME/examples/jars/spark-examples_2.12-3.5.5.jar 100
25/06/06 18:06:04 INFO SparkContext: Running Spark version 3.5.5
25/06/06 18:06:04 INFO SparkContext: OS info Linux, 6.8.0-60-generic, amd64
25/06/06 18:06:04 INFO SparkContext: Java version 1.8.0_452
25/06/06 18:06:04 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
25/06/06 18:06:04 INFO ResourceUtils: ==============================================================
25/06/06 18:06:04 INFO ResourceUtils: No custom resources configured for spark.driver.
25/06/06 18:06:04 INFO ResourceUtils: ==============================================================
25/06/06 18:06:04 INFO SparkContext: Submitted application: Spark Pi
25/06/06 18:06:04 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, a
mount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount
: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
25/06/06 18:06:05 INFO ResourceProfile: Limiting resource is cpus at 1 tasks per executor
25/06/06 18:06:05 INFO ResourceProfileManager: Added ResourceProfile id: 0
25/06/06 18:06:05 INFO SecurityManager: Changing view acls to: hduser
25/06/06 18:06:05 INFO SecurityManager: Changing modify acls to: hduser
25/06/06 18:06:05 INFO SecurityManager: Changing view acls groups to:
25/06/06 18:06:05 INFO SecurityManager: Changing modify acls groups to:
25/06/06 18:06:05 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view perm
issions: hduser; groups with view permissions: EMPTY; users with modify permissions: hduser; groups with modify permissi
ons: EMPTY
25/06/06 18:06:06 INFO Utils: Successfully started service 'sparkDriver' on port 44869.
```

```
Windows PowerShell   hduser@dist: /media/   hduser@dist: ~/spark   hduser@dist: ~   hduser@dist: ~
25/06/06 18:07:40 INFO TaskSetManager: Starting task 99.0 in stage 0.0 (TID 99) (dist, executor 2, partition 99, PROCESS
_LOCAL, 9315 bytes)
25/06/06 18:07:40 INFO TaskSetManager: Finished task 98.0 in stage 0.0 (TID 98) in 51 ms on dist (executor 2) (98/100)
25/06/06 18:07:40 INFO TaskSetManager: Finished task 97.0 in stage 0.0 (TID 97) in 80 ms on dist (executor 1) (99/100)
25/06/06 18:07:40 INFO TaskSetManager: Finished task 99.0 in stage 0.0 (TID 99) in 43 ms on dist (executor 2) (100/100)
25/06/06 18:07:40 INFO DAGScheduler: ResultStage 0 (reduce at SparkPi.scala:38) finished in 55.406 s
25/06/06 18:07:40 INFO YarnScheduler: Removed TaskSet 0.0, whose tasks have all completed, from pool
25/06/06 18:07:40 INFO DAGScheduler: Job 0 is finished. Cancelling potential speculative or zombie tasks for this job
25/06/06 18:07:40 INFO YarnScheduler: Killing all running tasks in stage 0: Stage finished
25/06/06 18:07:41 INFO DAGScheduler: Job 0 finished: reduce at SparkPi.scala:38, took 56.043466 s
Pi is roughly 3.1416051141605115
25/06/06 18:07:41 INFO SparkContext: SparkContext is stopping with exitCode 0.
25/06/06 18:07:41 INFO SparkUI: Stopped Spark web UI at http://dist:4040
25/06/06 18:07:41 INFO YarnClientSchedulerBackend: Interrupting monitor thread
25/06/06 18:07:41 INFO YarnClientSchedulerBackend: Shutting down all executors
25/06/06 18:07:41 INFO YarnSchedulerBackend$YarnDriverEndpoint: Asking each executor to shut down
25/06/06 18:07:41 INFO YarnClientSchedulerBackend: YARN client scheduler backend Stopped
25/06/06 18:07:41 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
25/06/06 18:07:41 INFO MemoryStore: MemoryStore cleared
25/06/06 18:07:41 INFO BlockManager: BlockManager stopped
25/06/06 18:07:41 INFO BlockManagerMaster: BlockManagerMaster stopped
25/06/06 18:07:41 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!
25/06/06 18:07:41 INFO SparkContext: Successfully stopped SparkContext
25/06/06 18:07:41 INFO ShutdownHookManager: Shutdown hook called
25/06/06 18:07:41 INFO ShutdownHookManager: Deleting directory /tmp/spark-6bc81ace-fa03-4220-a1c5-dbfaae646085
25/06/06 18:07:41 INFO ShutdownHookManager: Deleting directory /tmp/spark-2bb38cbb-70a6-42a7-8250-100cdb4fbaa4
hduser@dist:~/spark-3.5.5-bin-hadoop3/examples/jars$
```

```
Windows PowerShell  hduser@dist: /media/  hduser@dist: ~/spark  hduser@dist: ~  dist
        at scala.concurrent.impl.CallbackRunnable.executeWithValue(Promise.scala:72)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete$1(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.$anonfun$tryComplete$1$adapted(Promise.scala:288)
        at scala.concurrent.impl.Promise$DefaultPromise.tryComplete(Promise.scala:288)
        at scala.concurrent.Promise.tryFailure(Promise.scala:112)
        at scala.concurrent.Promise.tryFailure$(Promise.scala:112)
        at scala.concurrent.impl.Promise$DefaultPromise.tryFailure(Promise.scala:187)
        at org.apache.spark.rpc.netty.NettyRpcEnv.org$apache$spark$rpc$netty$NettyRpcEnv$$onFailure$1(NettyRpcEnv.scala:
214)
        at org.apache.spark.rpc.netty.NettyRpcEnv$$anon$1.run(NettyRpcEnv.scala:264)
        at java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:511)
        at java.util.concurrent.FutureTask.run(FutureTask.java:266)
        at java.util.concurrent.ScheduledThreadPoolExecutor$ScheduledFutureTask.access$201(ScheduledThreadPoolExecutor.j
ava:180)
        at java.util.concurrent.ScheduledThreadPoolExecutor$ScheduledFutureTask.run(ScheduledThreadPoolExecutor.java:293
)
        at java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1149)
        at java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:624)
        at java.lang.Thread.run(Thread.java:750)
Caused by: java.util.concurrent.TimeoutException: Cannot receive any reply from /10.0.2.15:46608 in 120 seconds
        at org.apache.spark.rpc.netty.NettyRpcEnv$$anon$1.run(NettyRpcEnv.scala:265)
        ... 7 more
25/06/06 18:14:22 ERROR YarnScheduler: Lost executor 1 on dist: Executor Process Lost
25/06/06 18:14:22 INFO TaskSetManager: task 11.0 in stage 0.0 (TID 11) failed because while it was being computed, its e
xecutor exited for a reason unrelated to the task. Not counting this failure towards the maximum number of failures for
the task.
```

```
Windows PowerShell  hduser@dist: /media/  hduser@dist: ~/spark  hduser@dist: ~/data/  dist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.security.authorization.tables.on.storagehandlers does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.server2.show.operation.drilldown.link does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.txn.acid.metrics.reporting.interval does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.zookeeper.killquery.namespace does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.repl.copy.file.list.iterator.retry does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.optimize.topnkey.efficiency.threshold does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.server2.webui.explain.output does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.llap.plugin.rpc.port does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.lock.query.string.max.length does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.llap.io.track.cache.usage does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.server2.authentication.jwt.jwks.skip.ssl.cert does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.repl.retry.initial.delay does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.hook.proto.events.clean.freq does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.use.orc.codec.pool does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.query.results.cache.max.size does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.auto.convert.anti.join does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.strict.timestamp.conversion does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.repl.bootstrap.dump.open.txn.timeout does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.cardinality.preserving.join.optimization.factor does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.stats.max.num.stats does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
25/06/06 18:18:39 WARN HiveMetaStore: Location: file:/home/hduser/data/sqltest/spark-warehouse/kv_test specified for non
-external table:kv_test
Time taken: 1.413 seconds
spark-sql (default)> load data local inpath '../kv.txt' into table kv_test;
Time taken: 1.05 seconds
spark-sql (default)>
```

```
Windows PowerShell  hduser@dist: /media/  hduser@dist: ~/spark  hduser@dist: ~/data/  dist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.optimize.topnkey.efficiency.threshold does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.server2.webui.explain.output does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.llap.plugin.rpc.port does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.lock.query.string.max.length does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.llap.io.track.cache.usage does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.server2.authentication.jwt.jwks.skip.ssl.cert does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.repl.retry.initial.delay does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.hook.proto.events.clean.freq does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.use.orc.codec.pool does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.query.results.cache.max.size does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.auto.convert.anti.join does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.strict.timestamp.conversion does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.repl.bootstrap.dump.open.txn.timeout does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.cardinality.preserving.join.optimization.factor does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.stats.max.num.stats does not exist
25/06/06 18:18:39 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
25/06/06 18:18:39 WARN HiveMetaStore: Location: file:/home/hduser/data/sqltest/spark-warehouse/kv_test specified for non
-external table:kv_test
Time taken: 1.413 seconds
spark-sql (default)> load data local inpath '../kv.txt' into table kv_test;
Time taken: 1.05 seconds
spark-sql (default)> select * from kv_test;
1       val1
131     val131
23      val23
Time taken: 2.63 seconds, Fetched 3 row(s)
spark-sql (default)>
```

```
Windows PowerShell | hduser@dist: /media/ | hduser@dist: ~/spark | hduser@dist: ~/data/ | dist

   /___/ .__/\_,_/_/ /_/\_\   version 3.5.5
      /_/

Using Scala version 2.12.18 (OpenJDK 64-Bit Server VM, Java 1.8.0_452)
Type in expressions to have them evaluated.
Type :help for more information.

scala> import org.apache.spark.sql.SparkSession;
import org.apache.spark.sql.SparkSession

scala> val spark = SparkSession.builder()
spark: org.apache.spark.sql.SparkSession.Builder = org.apache.spark.sql.SparkSession$Builder@6573b345

scala>   .appName("Spark Hive Example")
res0: org.apache.spark.sql.SparkSession.Builder = org.apache.spark.sql.SparkSession$Builder@6573b345

scala>   .config("spark.sql.warehouse.dir","hdfs://localhost/user/hive/warehouse")
res1: org.apache.spark.sql.SparkSession.Builder = org.apache.spark.sql.SparkSession$Builder@6573b345

scala>   .enableHiveSupport()
res2: org.apache.spark.sql.SparkSession.Builder = org.apache.spark.sql.SparkSession$Builder@6573b345

scala>   .getOrCreate();
25/06/06 18:20:53 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
res3: org.apache.spark.sql.SparkSession = org.apache.spark.sql.SparkSession@666e1965

scala>
```

```
Windows PowerShell | hduser@dist: /media/ | hduser@dist: ~/spark | hduser@dist: ~/data/ | dist

scala> val spark = SparkSession.builder().appName("Spark Hive Example").config("spark.sql.warehouse.dir", "hdfs://localh
ost/user/hive/warehouse").enableHiveSupport().getOrCreate();
25/06/06 18:21:21 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
spark: org.apache.spark.sql.SparkSession = org.apache.spark.sql.SparkSession@666e1965

scala> spark.sql("show tables").show();
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.repl.dump.metadata.only.for.external.table does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.druid.rollup does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.repl.retain.prev.dump.dir does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.log.explain.output.to.console does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.server2.wm.delayed.move.timeout does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.privilege.synchronizer does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.metastore.wm.default.pool.size does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.llap.task.scheduler.preempt.independent does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.llap.io.cvb.memory.consumption. does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.llap.output.format.arrow does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.tez.llap.min.reducer.per.executor does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.llap.io.path.cache.size does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.arrow.root.allocator.limit does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.optimize.topnkey.max does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.splits.available.slots.calculator.class.name does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.vectorized.use.checked.expressions does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.optimize.delete.metadata.only does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.optimize.filter.preds.sort does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.compactor.wait.timeout does not exist
25/06/06 18:21:28 WARN HiveConf: HiveConf of name hive.llap.nodehealthchecks.executorratio does not exist
```

```
Windows PowerShell | hduser@dist: /media/ | hduser@dist: ~/spark | hduser@dist: ~/data/ | dist

25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.llap.plugin.rpc.port does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.lock.query.string.max.length does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.llap.io.track.cache.usage does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.server2.authentication.jwt.jwks.skip.ssl.cert does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.repl.retry.initial.delay does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.hook.proto.events.clean.freq does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.use.orc.codec.pool does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.query.results.cache.max.size does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.auto.convert.anti.join does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.strict.timestamp.conversion does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.repl.bootstrap.dump.open.txn.timeout does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.cardinality.preserving.join.optimization.factor does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.stats.max.num.stats does not exist
25/06/06 18:21:32 WARN HiveConf: HiveConf of name hive.mapjoin.full.outer does not exist
25/06/06 18:21:36 WARN ObjectStore: Version information not found in metastore. hive.metastore.schema.verification is no
t enabled so recording the schema version 2.3.0
25/06/06 18:21:36 WARN ObjectStore: setMetaStoreSchemaVersion called but recording version is disabled: version = 2.3.0,
 comment = Set by MetaStore hduser@10.0.2.15
25/06/06 18:21:37 WARN ObjectStore: Failed to get database global_temp, returning NoSuchObjectException
+---------+---------+-----------+
|namespace|tableName|isTemporary|
+---------+---------+-----------+
|  default|  kv_test|      false|
+---------+---------+-----------+


scala>
```

```
scala> hduser@dist:~/data/sqltest$ ^C
hduser@dist:~/data/sqltest$ spark-submit --master yarn --num-executors 3 --class org.apache.spark.examples.ml.JavaRandom
ForestClassifierExample $SPARK_HOME/examples/jars/spark-examples_2.12-3.5.5.jar
25/06/06 18:22:03 INFO SparkContext: Running Spark version 3.5.5
25/06/06 18:22:03 INFO SparkContext: OS info Linux, 6.8.0-60-generic, amd64
25/06/06 18:22:03 INFO SparkContext: Java version 1.8.0_452
25/06/06 18:22:03 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
25/06/06 18:22:04 INFO ResourceUtils: ==============================================================
25/06/06 18:22:04 INFO ResourceUtils: No custom resources configured for spark.driver.
25/06/06 18:22:04 INFO ResourceUtils: ==============================================================
25/06/06 18:22:04 INFO SparkContext: Submitted application: JavaRandomForestClassifierExample
25/06/06 18:22:04 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, a
mount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount
: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
25/06/06 18:22:04 INFO ResourceProfile: Limiting resource is cpus at 1 tasks per executor
25/06/06 18:22:04 INFO ResourceProfileManager: Added ResourceProfile id: 0
25/06/06 18:22:04 INFO SecurityManager: Changing view acls to: hduser
25/06/06 18:22:04 INFO SecurityManager: Changing modify acls to: hduser
25/06/06 18:22:04 INFO SecurityManager: Changing view acls groups to:
25/06/06 18:22:04 INFO SecurityManager: Changing modify acls groups to:
25/06/06 18:22:04 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view perm
issions: hduser; groups with view permissions: EMPTY; users with modify permissions: hduser; groups with modify permissi
ons: EMPTY
25/06/06 18:22:04 INFO Utils: Successfully started service 'sparkDriver' on port 36741.
```

```
25/06/06 18:22:14 INFO Client: Uploading resource file:/tmp/spark-51e0b37d-28dd-48fe-b31a-cb9df68db6c9/__spark_conf__157
8633452828014288.zip -> hdfs://localhost:9000/user/hduser/.sparkStaging/application_1749232395036_0007/__spark_conf__.zi
p
25/06/06 18:22:14 INFO SecurityManager: Changing view acls to: hduser
25/06/06 18:22:14 INFO SecurityManager: Changing modify acls to: hduser
25/06/06 18:22:14 INFO SecurityManager: Changing view acls groups to:
25/06/06 18:22:15 INFO SecurityManager: Changing modify acls groups to:
25/06/06 18:22:15 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view perm
issions: hduser; groups with view permissions: EMPTY; users with modify permissions: hduser; groups with modify permissi
ons: EMPTY
25/06/06 18:22:15 INFO Client: Submitting application application_1749232395036_0007 to ResourceManager
25/06/06 18:22:15 INFO YarnClientImpl: Submitted application application_1749232395036_0007
25/06/06 18:22:16 INFO Client: Application report for application_1749232395036_0007 (state: ACCEPTED)
25/06/06 18:22:16 INFO Client:
         client token: N/A
         diagnostics: [Fri Jun 06 18:22:15 +0000 2025] Application is added to the scheduler and is not yet activated. S
kipping AM assignment as cluster resource is empty.  Details : AM Partition = <DEFAULT_PARTITION>; AM Resource Request =
 <memory:1024, vCores:1>; Queue Resource Limit for AM = <memory:0, vCores:0>; User AM Resource Limit of the queue = <mem
ory:0, vCores:0>; Queue AM Resource Usage = <memory:1024, vCores:1>;
         ApplicationMaster host: N/A
         ApplicationMaster RPC port: -1
         queue: root.default
         start time: 1749234135067
         final status: UNDEFINED
         tracking URL: http://dist:50000/proxy/application_1749232395036_0007/
         user: hduser
```

```
        at org.apache.spark.deploy.SparkSubmit.main(SparkSubmit.scala)
hduser@dist:~$ source ~/.bashrc
hduser@dist:~$ spark-submit --master yarn --num-executors 3 --class org.apache.spark.examples.ml.JavaALSExample $SPARK_H
OME/examples/jars/spark-examples_2.12-3.5.5.jar > out_als
25/06/06 18:22:59 INFO SparkContext: Running Spark version 3.5.5
25/06/06 18:22:59 INFO SparkContext: OS info Linux, 6.8.0-60-generic, amd64
25/06/06 18:22:59 INFO SparkContext: Java version 1.8.0_452
25/06/06 18:22:59 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
25/06/06 18:23:00 INFO ResourceUtils: ==============================================================
25/06/06 18:23:00 INFO ResourceUtils: No custom resources configured for spark.driver.
25/06/06 18:23:00 INFO ResourceUtils: ==============================================================
25/06/06 18:23:00 INFO SparkContext: Submitted application: JavaALSExample
25/06/06 18:23:00 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, a
mount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount
: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
25/06/06 18:23:00 INFO ResourceProfile: Limiting resource is cpus at 1 tasks per executor
25/06/06 18:23:00 INFO ResourceProfileManager: Added ResourceProfile id: 0
25/06/06 18:23:00 INFO SecurityManager: Changing view acls to: hduser
25/06/06 18:23:00 INFO SecurityManager: Changing modify acls to: hduser
25/06/06 18:23:00 INFO SecurityManager: Changing view acls groups to:
25/06/06 18:23:00 INFO SecurityManager: Changing modify acls groups to:
25/06/06 18:23:00 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view perm
issions: hduser; groups with view permissions: EMPTY; users with modify permissions: hduser; groups with modify permissi
ons: EMPTY
25/06/06 18:23:00 INFO Utils: Successfully started service 'sparkDriver' on port 36619.
25/06/06 18:23:00 INFO SparkEnv: Registering MapOutputTracker
```

```
         diagnostics: [Fri Jun 06 18:16:03 +0000 2025] Application is added to the scheduler and is not yet activated. Q
ueue's AM resource limit exceeded.  Details : AM Partition = <DEFAULT_PARTITION>; AM Resource Request = <memory:1024, vC
ores:1>; Queue Resource Limit for AM = <memory:1024, vCores:1>; User AM Resource Limit of the queue = <memory:1024, vCor
es:1>; Queue AM Resource Usage = <memory:1024, vCores:1>;
         ApplicationMaster host: N/A
         ApplicationMaster RPC port: -1
         queue: root.default
         start time: 1749233763744
         final status: UNDEFINED
         tracking URL: http://dist:50000/proxy/application_1749232395036_0006/
         user: hduser
25/06/06 18:16:35 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:17:05 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:17:35 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:18:06 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:18:36 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:19:07 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:19:37 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:20:07 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:20:37 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:21:08 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:21:38 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:22:08 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:22:39 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:23:09 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
25/06/06 18:23:39 INFO Client: Application report for application_1749232395036_0006 (state: ACCEPTED)
```

```
25/06/06 18:22:14 INFO SecurityManager: Changing view acls to: hduser
25/06/06 18:22:14 INFO SecurityManager: Changing modify acls to: hduser
25/06/06 18:22:14 INFO SecurityManager: Changing view acls groups to:
25/06/06 18:22:15 INFO SecurityManager: Changing modify acls groups to:
25/06/06 18:22:15 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view perm
issions: hduser; groups with view permissions: EMPTY; users with modify permissions: hduser; groups with modify permissi
ons: EMPTY
25/06/06 18:22:15 INFO Client: Submitting application application_1749232395036_0007 to ResourceManager
25/06/06 18:22:15 INFO YarnClientImpl: Submitted application application_1749232395036_0007
25/06/06 18:22:16 INFO Client: Application report for application_1749232395036_0007 (state: ACCEPTED)
25/06/06 18:22:16 INFO Client:
         client token: N/A
         diagnostics: [Fri Jun 06 18:22:15 +0000 2025] Application is added to the scheduler and is not yet activated. S
kipping AM assignment as cluster resource is empty.  Details : AM Partition = <DEFAULT_PARTITION>; AM Resource Request =
 <memory:1024, vCores:1>; Queue Resource Limit for AM = <memory:0, vCores:0>; User AM Resource Limit of the queue = <mem
ory:0, vCores:0>; Queue AM Resource Usage = <memory:1024, vCores:1>;
         ApplicationMaster host: N/A
         ApplicationMaster RPC port: -1
         queue: root.default
         start time: 1749234135067
         final status: UNDEFINED
         tracking URL: http://dist:50000/proxy/application_1749232395036_0007/
         user: hduser
25/06/06 18:22:46 INFO Client: Application report for application_1749232395036_0007 (state: ACCEPTED)
25/06/06 18:23:16 INFO Client: Application report for application_1749232395036_0007 (state: ACCEPTED)
25/06/06 18:23:46 INFO Client: Application report for application_1749232395036_0007 (state: ACCEPTED)
```